

Octopus AI — Technical Specification

Overview

Octopus AI is a multi-module autonomous platform for sales & marketing automation. This document details the technical capabilities, data formats, and integration protocols.

Platform Stack

Layer	Technology
Frontend	React 18, Tailwind CSS, Framer Motion
Backend	API Routes (serverless functions)
Database	PostgreSQL with Prisma ORM
Authentication	NextAuth.js (credentials + OAuth)
AI Models	Multi-provider (OpenAI, Anthropic Claude, Google Gemini, ElevenLabs, Kling, Veo)
Real-Time	Server-Sent Events (streaming responses)
Storage	AWS S3 (cloud file storage)
Payments	Stripe (subscriptions + usage billing)
Deployment	Serverless (standalone build)

OCTOPUS Agent System

Prompt Architecture

The OCTOPUS agent uses a **3-layer modular prompt system**:

- Core Personality (~3K tokens)** — Always injected. Defines identity, tone, capabilities, and behavioral rules.
- Module Contexts** — Selectively injected by the **Context Router** based on user intent. Each module (Growth Engine, Code Engine, etc.) has its own context block with specific instructions.
- Action Contexts** — Action examples and response templates for active modules.

Context Router

The Context Router (`context-router.ts`) analyzes user messages via keyword matching and injects only relevant module contexts. Each module defines:

- **Signals**: Keywords that activate the module (e.g., “lead”, “campaign”, “prospecting” → Growth En-

gine)
- **Companions:** Related modules that are co-injected for cross-module workflows

RAG 2.0+ (5-Layer Retrieval)

Layer	Source	Purpose
1. Semantic Memory	Conversation history	Short-term context and user intent continuity
2. Knowledge Base	User-uploaded documents	Domain-specific knowledge retrieval
3. Vector Search	Embeddings store	Semantic similarity matching across all data
4. Graph Context	Relationship mapping	Entity relationships and cross-module connections
5. Arms (Brazos)	External integrations	Real-time data from GitHub, Gmail, APIs, etc.

Global State System

The `buildGlobalState(userId)` function executes ~19 parallel database queries to build a comprehensive snapshot:

- User plan + usage limits
- Connected Brazos (integrations) with configuration details
- Active voice agents, sales agents (with per-agent lead counts)
- IoT devices and their states
- Growth Engine campaigns with metrics
- Creative assets, social bridge connections
- Invoices, calendar events
- Knowledge base documents
- Subscription status

This state is always injected into the agent prompt, giving OCTOPUS real-time awareness of the user’s entire workspace.

Module Specifications

Growth Engine

- **Lead Scoring:** AI-powered scoring based on engagement signals, profile completeness, and interaction history
- **Campaign Types:** Email sequences, social outreach, ad campaigns, content funnels
- **Prospecting:** Automated lead discovery with filtering by industry, location, company size
- **Actions:** `create_campaign` , `assign_leads_to_campaign` , `generate_prospects` , `lead_scoring`

Code Engine

- **AI Model:** Claude Sonnet 4.6 (default), supports model selection
- **Bridge System:** Octopus Bridge — local desktop app that syncs files between the platform and user's machine
- **Templates:** 6 built-in (3 Frontend: SaaS Dark, Portfolio, E-commerce; 3 Full-Stack: SaaS Dashboard, Blog CMS, CRM Pipeline)
- **Deploy Targets:** GitHub, Vercel, Netlify, Railway, Hostinger, Octopus Pages, ZIP export
- **Features:** Real-time preview (iframe with Babel transpilation), Git integration (push/pull/branches/commits), AI code review (score 0-100), workspace intelligence (file tracking in DB)
- **Prompt System:** 17 sections (§1-§17) with dynamic schema injection from Prisma models

Creative Studio

- **Asset Types:** Images (DALL-E, Flux), Videos (Veo 3.1, Kling, Seedance), Audio (ElevenLabs TTS), Documents (PDF generation)
- **Lead-to-Asset Pipeline:** Automatically generates personalized assets per lead (video with name, company, custom messaging)
- **UGC Factory:** AI-generated user-generated content style videos

Sales Agent

- **Embeddable:** JavaScript widget for any website
- **Customizable:** Custom personality, knowledge base, appearance
- **Lead Capture:** Real-time form collection during conversations
- **Analytics:** Conversation metrics, conversion tracking

Voice Agent

- **TTS Providers:** ElevenLabs (premium voices), browser TTS (fallback)
- **STT:** Real-time speech-to-text
- **Storage:** Database-backed (VoiceAgent model with full CRUD)
- **Integration:** Works with Sales Agent for voice-enabled chat

Integration Protocols (Brazos)

GitHub Integration

- **Auth:** OAuth (personal access token)
- **Capabilities:** Push/Pull repos, list/create/merge branches, list commits, AI code review
- **API:** Git Trees API (recursive), capped at 200 files, 500KB per file

Gmail Integration

- **Auth:** OAuth 2.0 (scope: gmail.send, gmail.readonly)
- **Actions:** `list_emails`, `get_message`, `send_email`, `create_draft`, `reply`, `search`
- **Safety:** OCTOPUS confirms with user before sending; drafts as fallback

Ollama Integration

- **Protocol:** HTTP API to local Ollama instance
- **Models:** Llama 3, Mistral, DeepSeek, and any Ollama-supported model
- **Use Case:** Private/local AI processing without cloud dependency

Data Models

Key database entities:

- **User** — Authentication, plan, preferences
- **ArmConnection** — External service connections (type, credentials, status)
- **CustomAgent** — User-created AI agents (name, personality, model, category)
- **CustomSkill** — User-created reusable AI skills
- **CustomMcp** — User-configured MCP (Model Context Protocol) servers
- **VoiceAgent** — Voice agent configurations
- **SalesAgent** — Sales agent configurations with lead tracking
- **OctoBlogPost** — Blog articles with SEO metadata
- **TaskItem** — Task management items
- **WorkspaceIndex** — Code Engine file tracking
- **SessionMemory** — Cross-session context for Code Engine

API Structure

```

/api/
├── jarvis/
│   ├── chat/                # Main OCTOPUS chat endpoint (streaming)
│   └── web-search/          # Web search with grounding
├── brazos/
│   ├── google/              # Gmail, Calendar, Drive
│   └── [service]/            # Other integration endpoints
├── agent-factory/
│   └── agents/               # CRUD for custom agents
├── skill-factory/
│   └── skills/               # CRUD for custom skills
├── mcp-factory/
│   └── servers/              # CRUD for MCP servers
├── voice-agent/              # CRUD for voice agents
├── sales-agent/              # CRUD for sales agents
├── growth/                   # Growth engine campaigns & leads
├── tasks/                    # Task management
├── blog/                     # Blog articles
├── arms/
│   └── claude-code/          # Code Engine (chat, GitHub, workspace, review)

```

Internationalization

- **Supported Languages:** Spanish (es), English (en)
- **Implementation:** Client-side locale detection + i18n dictionary
- **Coverage:** All UI elements, agent responses, error messages, metadata